

Group Title: The Glass Canvas: How Screens Evolved

Group Members (Group 7):

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Group Theme:

Understanding different display technologies and how they produce images on screen.

Group Tech Stack:

- Astro: Project structure skeleton.
- React: For interactive components and UI render.
- MDX: Dynamic elements.
- React Three Fiber: For handling 3d rendering and interactions.
- Drei: Library containing premade helper functions for React Three Fiber.

Project Description:

An interactive educational platform designed to help users understand the different types of displays developed and how its components work together to deliver visuals to us. The platform covers key components such as why display screens were developed, the uses of each display type, and comparing them to each other. Users can explore detailed explanations, interactive visualizations, and dynamic content that demonstrate the role of each component and how the components work together to produce light and process it to produce images. The project is built using Astro for the application structure, React for interactive user interfaces, MDX for content-driven learning experiences, React Three Fiber (R3F) for 3d models, and Drei as a collection of helper functions for R3F.

Exhibit Navigation will be composed of the following pages:

1. **Rationale:** Discusses why display technology was developed and what needs did it address.
2. **Timeline:** A user will be presented with a long scroll timeline to choose which display tech they wish to learn more about. Each entry in the timeline will provide a year on when the screen became commercially available to the public along with a high-level overview on how each display type works and their applications.
3. **The Cathode Ray Tube:** Discuss what a cathode ray tube is, its parts, and how it works. This section will also show its advantages and limitations, along with its applications in the past and present.
4. **Plasma:** Similar to the Cathode Ray Tube, its origins, applications, and inner workings will be discussed. Will also cover its early and late use during its era along with its decline.
5. **LCD Family:** Will mainly focus on its core principles of a liquid crystal layer, having two polarizing light filters, susceptibility to light bleed, power delivery, backlighting, and how the other variations either replace some of the components or build on top of it.
6. **OLED Family:** Exploring its application in consumer electronics, IoT, embedded systems, industrial, and automotive.
7. **MicroLED:** Currently an in development display screen type. Combines the strengths of LCD and OLED type screens.

Interactive Element: (3d rendering with React Three Fiber and Drei)

- **3D Component Explorer:** A 3D model to be used for showcasing the different components for LCD and OLED displays.
 - Component descriptions will be displayed accordingly depending on the component the user clicks on the 3D model.
 - Users can rotate the 3D model.
 - Shows animations to simulate how light passes from its source and is processed into colors.

- The model should automatically zoom in to the component the user wishes to know more about.

Exhibition Content:

1. Introduction

- How and Why Display Technology Rose in the First Place

2. Cathode Ray Tube (CRT)

a. Introduce CRT

- i. Brief explanation of CRT
- ii. Who developed the CRT

b. Parts of a CRT

c. How a CRT works

d. Applications of CRT

- i. Where CRT is being used when it was first introduced
- ii. Where CRT is being used currently

e. Advantages and Limitations of CRT

3. Plasma

a. Introduction and Origin of Plasma Display Panel

b. Parts of Plasma Display

c. How Plasma Works

d. Applications of Plasma

- i. Early use in computer display
- ii. Later use in consumer homes

e. Advantages and Limitations of Plasma

f. Decline in Use

4. LCD Family

a. What are LCD Displays?

- i. High-level definition
- ii. Uses of LCD
- iii. Applications of LCD

b. LCD Principles

- i. Liquid Crystal

- ii. Backlighting

- 1. Cold Cathode Fluorescent Lamp (CCFL)

- iii. Light Polarization

- c. The Upgrade to LED

- i. Light Emitting Diode (LED) backlight vs CCFL

- ii. Direct vs Edge Backlighting for LED.

- iii. Blooming Effect Problems of Both CCFL and LED

- d. Moving into Mini-LED

- i. Full-array Local Dimming (FALD)

- ii. Mitigating Bloom

- e. Topping it off with QDLED

- i. Introduction to the Quantum Dot Layer

- ii. How it's used in LCD Displays.

5. OLED

- a. Introduction and Fundamentals of OLED

- b. OLED Components

- c. How OLED works

- d. Applications of OLED

- i. Consumer Electronics

- ii. Automotive Systems

- iii. Industrial and Embedded Systems

- iv. IoT devices

- e. OLED Evaluation

- i. Advantages

- ii. Limitations

6. QDOLED

- a. Introduction and Fundamentals of QD-OLED

- b. QD-OLED Components

- c. How QD-OLED works

- d. Applications of QD-OLED

- i. Gaming Monitors

- ii. Premium Tvs
 - e. OLED Evaluation
 - i. Advantages
 - ii. Limitations
 - f. Compare and Contrast with OLED
- 7. MicroLED
 - a. Introduction and Fundamentals of MicroLED
 - b. How MicroLED works
 - c. Strengths and Limitations of MicroLED
 - d. Discussion on how MicroLED possesses the strengths of both LCD and OLED.

Tentative Style Guide Snapshot:

Whenever a user clicks the button from the timeline to redirect to a 3D model or a 2D image, this page will appear and provide descriptions per component. 3D models will get zoom-ins on the selected component while 2D images will highlight.

The Glass Canvas: How Screens Evolved	
Section here discusses how and why display technology was developed. Includes what issues it addresses and the other benefits it brought to society.	
	CRT - Year of Commercial Availability Insert text here discussing the application of CRT displays. Both current and historical. Include who developed the CRT. Go To 2D Image
	Plasma - Year of Commercial Availability Provide High-level Overview on what are plasma displays and how they work. Discuss their origins and applications. Go To 2D Image
	LCD Family- Year of Commercial Availability High level overview of LCD technology, briefly discuss it's other variations: LED, QLED, Mini-LED. Discuss Applications of LCD. Go To 3D Model
	OLED Family - Year of Commercial Availability High Level overview of OLED technology, briefly discuss it's other variation: QOOLED. Go To 3D Model
	MicroLED - Year of Commercial Availability High Level overview of MicroLED. The current frontier in display tech. It's advantages over the preceding display types. Go to 2D image

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